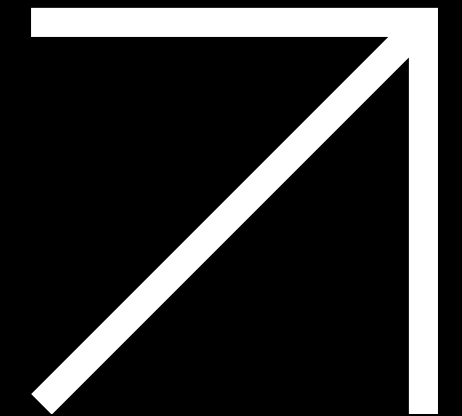




Karl-Franzens-Universität Graz

Veritatem

AI-Based Clinical Decision Support for Early GDM Risk



Course
Economic Aspects of AI

Members
Satera Sleinyte, Julian Baron, Žan Porenta

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What is Veritatem?



Veritatem is an AI-based clinical decision-support tool that estimates gestational diabetes risk in the first trimester using routine antenatal data. It gives clinicians a risk score, an explanation, and a suggested next step while keeping the clinician in control.

Early risk detection

Uses first-trimester data such as age, BMI, medical history and lab values

Clinical Interpretation

Shows which factors influenced the risk score, so the result is understandable

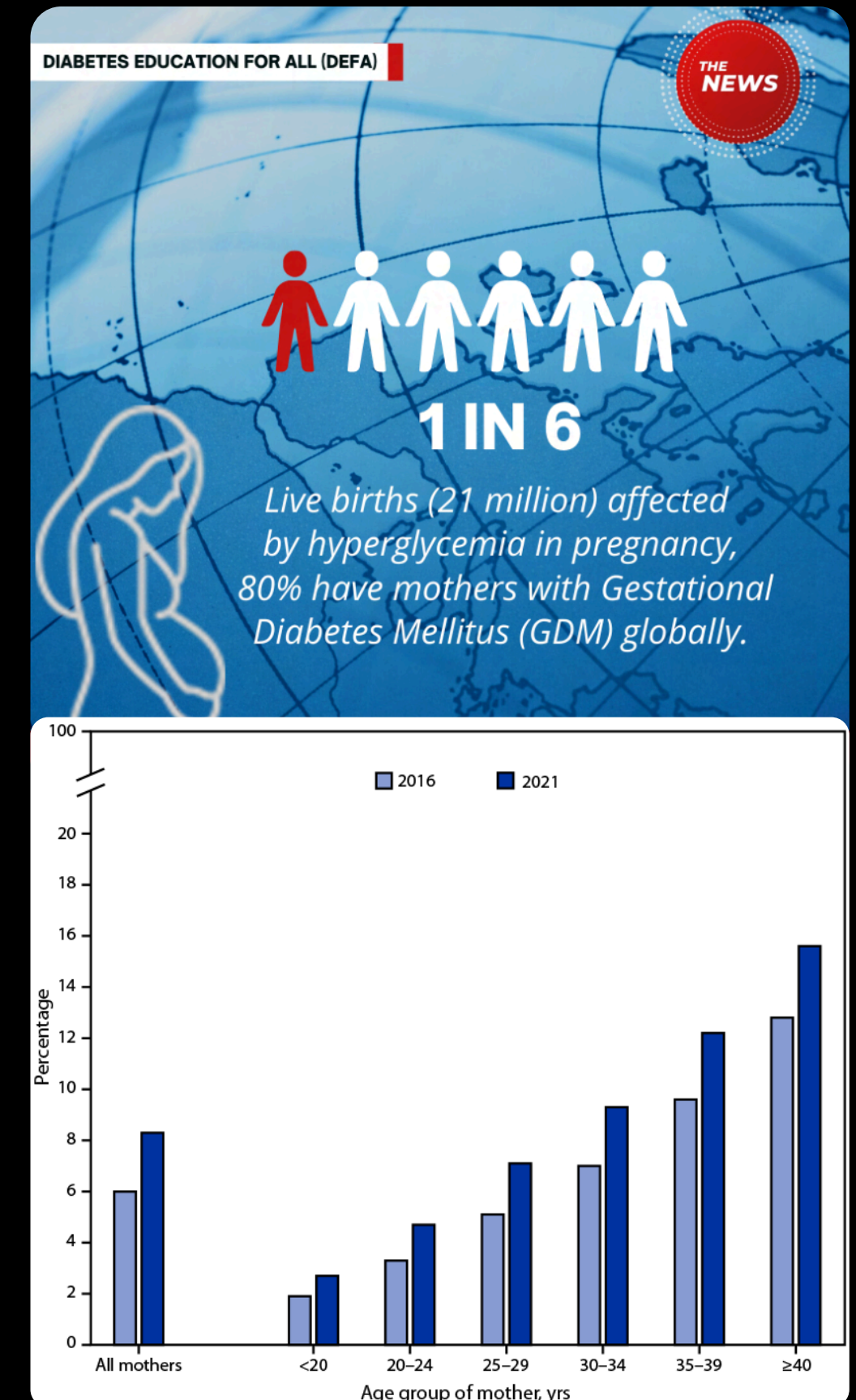
Decision support

Suggests next steps such as early OGTT, lifestyle referral or closer monitoring

Why timing matters...

- GDM is high blood sugar first recognised during pregnancy
- Affects around 11% of pregnancies in Europe
- Linked to maternal and neonatal complications
- Standard OGTT testing usually happens at 24–28 weeks
- By then, the window for early lifestyle support and monitoring may be smaller

Late diagnosis limits the time clinicians have to act. The earlier the risk is visible, the earlier care can shift from reaction to prevention. In other words: earlier insight means more time for low-cost prevention.



1. <https://www.diabetesdefa.org/index.php/assets/image/global-impact-1-6-live-births-affected-hyperglycemia-pregnancy>

2. <https://www.cdc.gov/mmwr/volumes/72/wr/mm7201a4.htm>

How does it support clinicians?



Veritatem is used during the first antenatal visit to turn routine patient data into an actionable risk profile. The goal is not to simply diagnose GDM, but to help clinicians decide whether earlier testing or support is needed.

Patient Profile

Age, BMI, previous GDM,
family history, early lab
values

Risk Profile

Low / moderate / high risk,
with probability and key
contributing factors

Action Pathway

Early OGTT, lifestyle
referral, closer monitoring,
or standard care

/// AI Model and Clinical Interpretation

AI for safe clinical use

01.

Tabular Prediction

Uses structured first trimester patient data such as age, BMI, previous GDM, family diabetes history and early lab values, instead of complex images or free-text records.

02.

Efficient Model Type

Gradient-boosted decision trees fit this problem because they work well with mixed clinical variables and missing values.

03.

Calibrated Risk Output

Gives a probability and risk tier, with uncertainty rather than a fake “exact” answer.

04.

Controlled Updates

The model is version-locked, logged and updated only through a formal process.

Clinician Dashboard and Workflow



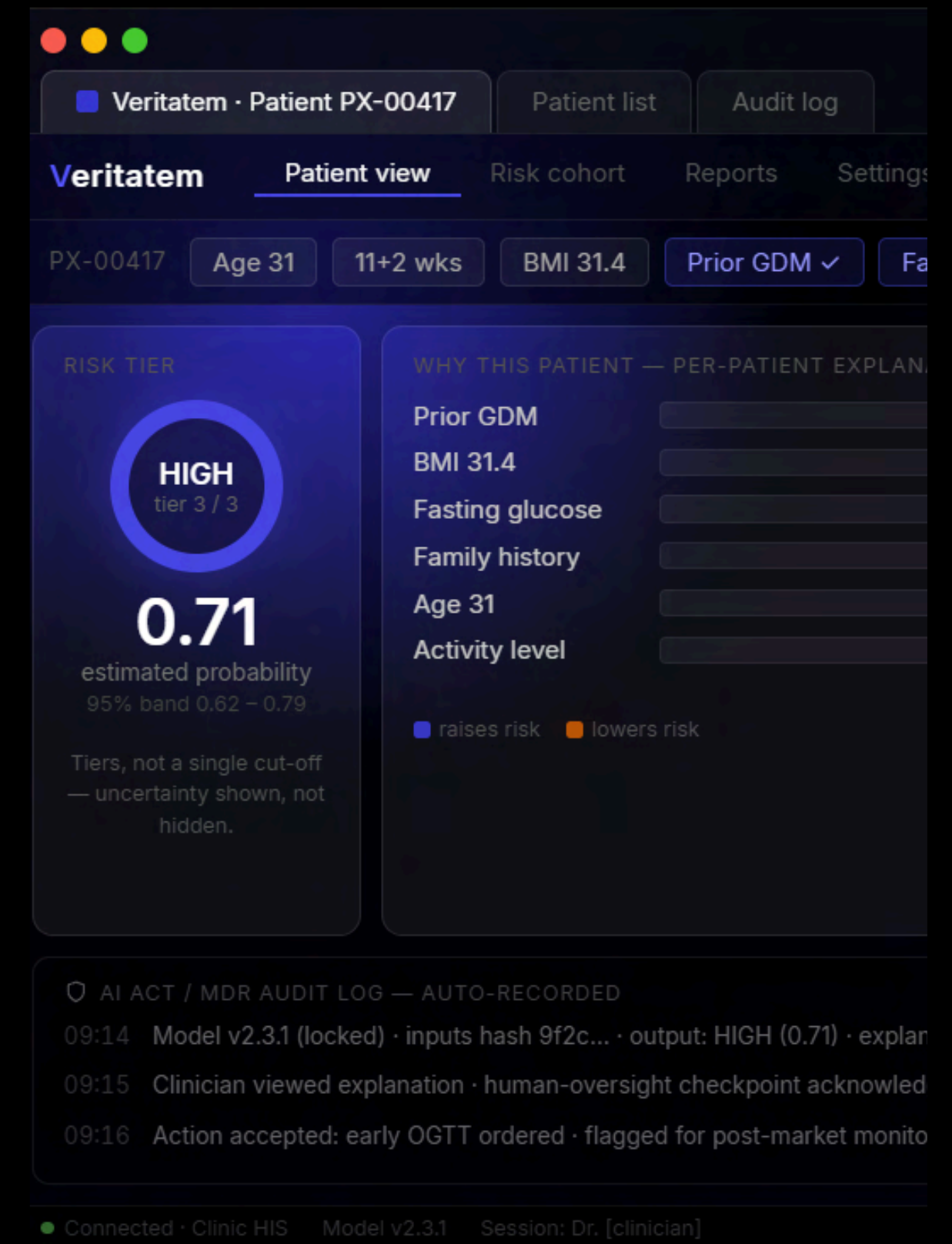
Competitive advantage?

More than the algorithm.

Veritatem's advantage is not the prediction model alone. Similar GDM risk models already exist in academic research. What is harder to copy is the full clinical product: local validation data, CE marking, AI Act compliance, explainable outputs, audit trails and integration into real hospital workflows.

- local validation makes the model credible for Austrian and Slovenian patients
- CE marking and AI Act compliance make it legally usable in EU healthcare
- audit trails and explanations make the system easier for clinicians to trust
- EHR integration makes adoption easier inside existing hospital workflows

The moat is the validated, regulated and trusted product - not just the AI model



Competitor analysis

Alternative	What it is	Strengths	Where Veritatem differs
Status quo (Oral Glucose Tolerance Test at 24–28 wks)	Standard guideline-based testing	Free, trusted, no friction, universal	Acts months earlier; flags risk while cheap interventions still have time to work
Published academic models	ML risk models in the literature	Often free, peer-reviewed, accurate in-sample	Not products: no CE mark, no integration, no local validation, no support or audit trail
EHR-native risk scores	Risk flags built into hospital EHRs	Already in the workflow; low marginal cost	Usually generic, rarely explainable, rarely validated locally or cleared as a device for this purpose
Commercial maternal-health AI	Vendor platforms / monitoring suites	Funded, polished, sales reach	Broad rather than GDM-early-window specific; explainability and local validation vary

Market Opportunity

Layer	Definition & method	Volume (approx.)	Indicative annual value
TAM	Total Addressable Market: All EU pregnancies passing through antenatal care that could in principle be screened (EU live births).	~3.55 million screenings / yr	~€35 million / yr at €10/screen
SAM	Serviceable Available Market: EU markets with structured antenatal care, feasible EHR integration, and a realistic reimbursement path - a subset of ~5–6 countries.	~1.5 million screenings / yr (assumption)	~€15 million / yr at €10/screen
SOM	Serviceable Obtainable Market: Austria–Slovenia beachhead, with realistic penetration over ~5 years at 15–30% of beachhead volume.	~95,000 births/yr beachhead → ~14,000–29,000 screenings/yr captured	~€140k–€290k / yr at maturity

The market opportunity is broad at EU level, but the realistic entry market is much smaller. The Austria–Slovenia beachhead is not large enough to make Veritatem highly profitable on its own. its main value is to generate validation data, test the clinical workflow and create first reference customers.

The long-term business case depends on expanding into larger EU markets and later securing payer support.

The revenue model?



Veritatem would begin with provider subscriptions and pay-per-screening fees because these are easier to sell before long-term outcome data exists. Larger payer contracts and EHR licensing would come later, once pilots prove clinical and economic value.

Pilots create evidence. Evidence creates trust. Trust enables scale.

Provider SaaS

Maternity units and antenatal clinics pay recurring fees to use the tool in their workflow.

Pay-per-Screening

Clinics pay per risk assessment, lowering the entry barrier for smaller providers.

Future Scale

Payer contracts and EHR licensing become realistic after pilots prove clinical and economic value.

The market strategy?



Veritatem cannot rely on mass marketing because clinical AI adoption depends on trust, evidence and workflow fit. The first step is to run pilots with maternity units, use the results to build credibility, and then expand through clinical champions, EHR platforms and payer-facing evidence.

First prove the product in clinics, then scale through evidence-based channels.

Pilot Projects

Test the tool in maternity units, generate local validation data and prove workflow fit.

Clinical Champions

Respected clinicians support adoption inside hospitals and make the product more credible.

Evidence Channels

Use EHR marketplaces, conferences, publications and HTA dossiers to support wider expansion.

Regulation and Operations.



Veritatem operates in a highly regulated healthcare environment. Compliance, data protection, and continuous monitoring are essential to ensure safe and trustworthy AI-supported clinical decisions.

Data & Partnerships

- EHR integration
- Secure clinical data pipeline
- Partnerships with clinics and universities

Regulatory Compliance

- EU MDR (Class IIa) Rule 11
- EU AI Act (High-Risk AI) Article 6 (1)
- GDPR Article 9

Quality & Monitoring

- ISO 13485, 14971 / IEC 62304
- Risk management
- Continuous performance monitoring

3€

Cost per Screening
(at 1M screenings)



Financial Logic and Scale-Up

Veritatem requires high upfront investment but benefits from strong economies of scale. Regulatory compliance and clinical validation drive costs, while marginal screening costs remain low.

Fixed costs

- Clinical validation
- Regulatory approval
- Quality management
- Development team

Marginal costs

- AI inference infrastructure
- Support
- EHR integration

Risks and Limitations.



Data Availability

No secured local training and validation data yet.



Clinical Evidence

Benefits of early detection are promising but not yet fully established.



Regulatory Uncertainty

MDR and AI Act requirements may evolve over time.



Scaling & Reimbursement

Multi-market expansion and reimbursement remain challenging.

 Conclusion /
Recommendation

Strong Long-Term
Potential.

01.

Data Access

Secure local training and
validation data.

02.

Clinical Validation

Establish clinical credibility
and evidence.

03.

Regulatory Approval

Comply with MDR, AI Act
and quality standards.

04.

Multi-Market Scale

Expand beyond the Austria–
Slovenia market.

Thank You.

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